AI Chat Usage Report

WASL Arena · Family AI Game Day · 6th Annual COE STEM Programming with AI Competition

Team 204 · Team code AYUCK · Track: Cycle 3 (grades 9-12)

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1. Purpose and authenticity statement

This report documents how the team used AI assistants to plan, build, test and document WASL Arena. Appendices A and B reproduce the **original conversation text exactly as exported**: Appendix A from the shared ChatGPT conversation, Appendix B from this Claude Code session (user messages and assistant replies). Nothing was rewritten or invented. The only change is **redaction of credentials** (API keys → [REDACTED-API-KEY]; 1 occurrence(s) redacted). Conversations that are not attached are not claimed.

2. AI tools and models used

Tool / model	Used for	Evidence
ChatGPT (OpenAI)	Concept ideation (WASL Arena, three games), consolidated master execution prompt, design-reference links (Spline, Recent.design)	Appendix A (share link in export)
Claude Code · Claude Sonnet 5.5 (Anthropic)	Implementation: backend, CV pipelines, rule engines, tests, React frontend, printables, report/deck/video/AI-report generators, debugging, QA	Appendix B (84 messages; 210 tool calls)
Gemini API · gemini-3.1-flash-lite (Google)	Runtime speech transcription in Souq Signal only (never scores). Model selected after the originally planned model was found retired for new users.	Evaluation Lab 'Speech transcription check'
Windows text-to-speech (SAPI)	Synthetic audio for test fixtures and scratch narration of the draft animatic	fixtures/, video/ (labelled)

3. Development phases and AI contribution

Phase	What the AI did	Human / verification step
1 Concept	Proposed WASL Arena and three games; consolidated one master prompt (ChatGPT)	Team chose the concept; brief read from the official question images
2 Architecture	Designed FastAPI + OpenCV + rule-engine architecture, evidence-integrity scheme (Claude Code)	Decisions checked against the brief (host flow, hash-verified reports, MVP = longest possession)
3 Code generation	Wrote rule engines, football/falaj/souq pipelines, API, React app (Claude Code)	Type-check and production build; UI inspected in a browser
4 Debugging	Found and fixed: well-inlet counted as leak; Arabic ال prefix not matched; accepted possible goal not scored; retired Gemini model	Failures surfaced by the test suite, then re-run
5 Testing	Wrote 46 tests and 9 synthetic fixtures with known ground truth	46/46 tests passed in the final run; fixtures are synthetic, not real gameplay
6 Design iterations	Applied the palette, Spline-inspired orb (original Three.js), Krea-style judging sequence; fixed contrast bug (invisible text) found in screenshots	Screenshots reviewed by the AI and team
7 Documentation	Generated the 6-page report, 30-slide deck, video package and this report from real results	Page count verified in Word (6 pages); slides rendered in PowerPoint (30)
8 Video scripting	Wrote the 24 × 5 s script: voiceover, image-generation prompts, animation prompts, shot list; built a draft animatic from real screenshots	Real gameplay clips (football, Falaj, Souq) must be recorded by the team; AI imagery limited to atmosphere clips
9 Deployment & debugging	Diagnosed the Render build failure from the team's pasted log (Root Directory not set), verified the live Render API with a real judging job, found the Vercel build lacked VITE_API_URL, verified CORS	Team created the Render and Vercel projects, set environment variables and redeployed; AI checked the live site afterwards

4. Human verification and adjustments

- The team supplied the official competition materials, team identity, the ChatGPT conversation and the speech API key; the key was kept in a git-ignored server-side `.env` and is redacted in this report. Because the key appeared in a chat, **rotate it after the competition**.
- All numbers in the report/deck/Evaluation Lab are read from files produced by running the tests and pipelines.
- Synthetic fixtures are always labelled; real footage is pending and is not represented as tested.
- Items still requiring human action: record real gameplay for football, Falaj and Souq, review final wording, record the voice-over and music, and rotate the API key used during development.

5. Claude Code activity summary

Tool category	Calls
Bash	80
Write	54
Read	19
Edit	13
ToolSearch	6
navigate	6
computer	6
javascript_tool	4
PowerShell	3
get_page_text	3
update_plan	2
WebFetch	2

Session window (UTC): 2026-10-29T11:20:55.336Z to 2026-10-29T12:45:30.619Z. The transcript was exported while work was in progress; later turns are not included.

Appendix A: ChatGPT conversation (verbatim export)

Source: https://chatgpt.com/share/6ae32cc9-fa3c-83eb-98ef-c07e7b8e125c · captured 2026-10-29T12:01:01+00:00 · 6 messages. Formatting flattened to text; file attachments are shown only as 'Uploaded a file' placeholders in the export.

USER · message 1

Uploaded a file

Uploaded a file

Uploaded an image

Uploaded a file

Generate specific prompts and ideas for the new prototype, following the question attached. Generate the best possible idea in accordance to the competition requirements.

CHATGPT · message 2

The refined concept is:

WASL Arena — Cycle 3 Edition

3 games

Football AI
Movement tracking

Falaj Flow
Visual + graph reasoning

Souq Signal
Vision + speech recognition

The three games exercise distinct AI capabilities, while satisfying the requirement for both team and individual play.

Final comprehensive master prompt

Paste the following into Claude along with the official ADU question and competition documents. It is designed to generate all five submission deliverables in a coordinated sequence.

MASTER PROMPT — WASL ARENA

MASTER PROMPT — WASL ARENA

ADU 6th Annual COE STEM Programming with AI Competition — Cycle 3

Act as an elite multidisciplinary team comprising:

-
Senior AI and computer vision engineer

-
Full-stack software architect

-

Python/FastAPI backend engineer

-
React/TypeScript frontend engineer

-
AI evaluation and quality assurance specialist

-
Premium UI/UX and creative director

-
Scientific and technical report writer

-
Competition presentation strategist

-
Cinematic video director and scriptwriter

-
GitHub, Vercel, and backend deployment specialist

Your assignment is to independently execute the development and submission-production workflow for WASL Arena — The Family AI Games Championship.

You must strictly adhere to the attached official Abu Dhabi University challenge brief and all submission requirements.

The target is the Cycle 3 Track (Grades 9–12), not the Advanced Champions Track.

There must be exactly three games: football plus two original games.

PART I — COMPETITION DEFINITION

1. Official challenge

Competition: 6th Annual COE STEM Programming with AI Competition

Theme: AI for Family

Problem: Family AI Game Day

Track: Cycle 3 — Grades 9–12

The challenge requires a working application that hosts family games inspired by culture, heritage or sports.

The human host sets up players and teams, explains rules, controls gameplay and manages rounds.

The AI observes actual gameplay through video and audio, detects relevant events, applies predefined rules and generates verifiable results.

2. Mandatory functionality

The solution must:

-
Include football and exactly two original games.

-
Support team and individual competition.

-
Allow families to choose their game.

-
Register players and teams.

-
Define points, timers, turns, rounds, penalties, ties and winners.

-
Accept uploaded gameplay videos with audio.

-
Support direct camera and microphone recording.

-
Analyze actual game footage.

-
Generate AI judgments grounded in observed evidence.

-
Explain how judgments were made.

-
Apply verified AI results to scores.

-
Display final scores, winners and MVP information where relevant.

-
Demonstrate actual gameplay and AI judging in the student video.

-
Provide a working, accessible, publicly deployed prototype.

Do not use the example original games shown in the problem statement.

The application must not depend on prerecorded animations or fabricated AI judgments.

PART II — THE CONCEPT

3. Product identity

Name: WASL Arena

Tagline: One family. Three games. One fair AI referee.

Core proposition: Transform family gaming into a connected experience where artificial intelligence watches, listens, judges and explains.

The innovation is the combination of:

-
Real physical games

-
Computer vision

-
Speech and audio recognition

-
Rule-based intelligence

-
Transparent scoring

-
UAE cultural relevance

-
Family interaction

-
Explainable AI

WASL Arena is not a collection of digital minigames.

The family plays in the real world, while the application operates as the game host, observer, referee and championship manager.

4. The signature innovation

AI REFEREE REPLAY

Every AI scoring decision must have an evidence trail.

For example:

Decision: Goal awarded to Team A.

Evidence: Video timestamp, detected player, tracked ball and observed goal-line crossing.

Rule: Football Rule F-03.

Score change: Team A +1.

Status: Verified or requires human review.

The user must be able to inspect the moment that produced the judgment.

Create an evidence timeline with playable video segments, frame annotations, detection outputs and explanations.

This is the centerpiece of the project's technical identity and competition presentation.

PART III — THE THREE GAMES

GAME 1 — FOOTBALL AI

Concept

A real family football match monitored by an AI referee.

Mode

Team versus team.

Default: Two players per team.

Features

-
Player registration

-
Team assignment

-
Team-color identification

-
Numbered player identification where feasible

-
Video recording and upload

-
Player and ball detection

-
Motion tracking

-
Goal detection

-
Score updates

-
Possession estimates

-
Match timer

-
Tie-break rules

-
MVP calculation

-
Evidence replay

Technical implementation

Use OpenCV, suitable object detection models and temporal tracking.

The initial demonstrable setup should use:

-
A clearly marked mini-football playing area

-
Two visible goals

-
Contrasting team bibs

-
Large readable player identifiers

-
A bright, visually detectable football

-
A stable camera angle

Analyze frames to locate players, the ball and calibrated goal regions.

Detect goal events by tracking the ball across the relevant goal boundary over consecutive frames.

Do not generate a goal merely because a language model describes one.

Compute a possession proxy using observed player/ball proximity when tracking data is reliable.

If a player, ball or goal event cannot be confidently distinguished, mark the event as unresolved.

Scoring

-
Verified goal: +1

-
Invalid goal: 0

-
Match winner: Highest score

-
Tie: Golden-goal extension or configured tie-break

-
MVP: Calculated from verified individual contributions only

Clearly distinguish measured possession from estimates.

GAME 2 — FALAJ FLOW

Concept

An AI-observed collaborative irrigation challenge inspired by traditional Emirati falaj systems.

Families compete to construct an efficient water distribution network.

Mode

Team versus team.

Physical setup

Use a printable 5×5 board with:

-
A water-source tile

-
Three agricultural destinations

-
Fixed obstacles

-
Connectable water-channel tiles

-
Distinctive printed visual markers

Players physically position and rotate channel tiles.

The objective

Connect the water source to as many destinations as possible while avoiding obstacles and invalid paths.

AI features

-
Board calibration

-
Tile identification

-
Tile orientation detection

-
Real-time or recorded board-state extraction

-
Route visualization

-
Graph construction

-
Connectivity verification

-
Invalid-path detection

-
Automatic scoring

Technical reasoning

Convert the observed tile arrangement into a graph.

Represent valid water connections as edges.

Use graph traversal to determine which agricultural destinations are reachable from the water source.

Reject paths blocked by obstacles or invalid tile orientation.

Scoring

-
Valid destination reached: +3

-
Invalid outlet or leak: -1

-
Completion-time bonus: Up to +2

-
Round duration: 90 seconds

-
Tie-break: Shortest valid network, then completion time

Visual wow factor

Display the physical board on one side and its AI-generated digital network on the other.

Animate verified water flow through the reconstructed graph.

Water must only flow through routes validated by actual observed placements.

GAME 3 — SOUQ SIGNAL

Concept

A family marketplace challenge combining object recognition, speech recognition, budgeting and logical reasoning.

The family receives a shopping mission and must complete it using physical item cards.

Mode

Individual competition, with optional alternating players.

Physical setup

Create a printable card collection containing:

- Dates
- Saffron
- Coffee
- Textiles
- Spices
- Pottery

Each item has:

- A unique visual identifier
- A fixed game price
- English and Arabic labels

Gameplay

The application generates a shopping mission.

Example:

"Purchase the dates, coffee and pottery within a budget of 15 dirhams."

The player selects cards, presents them to the camera and verbally announces the items.

The AI compares:

-
The requested shopping list

-
The visually observed cards

-
The recognized spoken selections

-
The total item cost

-
The remaining budget

Technical implementation

Use visual markers for reliable card identification.

Use speech transcription for spoken selections.

Match recognized words to a fixed bilingual item vocabulary.

Do not rely on open-ended language-model judgment to determine whether a purchase is valid.

Scoring

-
Correct requested item: +2

-
Verified agreement between spoken and visually observed item: +1

-
Budget exceeded: -2

-
Duplicate item: No additional item points

-
Tie-break: Lowest valid expenditure, then completion time

Visual wow factor

Show the live camera feed beside:

-
AI-recognized products

-
Speech transcript

-
Shopping basket

-
Budget calculation

-
Rule validation

-
Automated score explanation

Use a prominent visual indicator when audio and video evidence agree.

If either input is ambiguous, request review instead of issuing an unsupported penalty.

PART IV — PROTOTYPE DEVELOPMENT

5. Required screens

Develop the following eight fully connected screens.

01. Launch Experience

Create a visually captivating introduction.

Include:

-
WASL Arena branding

-
Three-game championship introduction

-
AI Referee positioning

-
Start Championship

-
Explore Games

-
How It Works

-
English/Arabic language selection

02. Family Registration

Allow:

-

Player registration

-
Player numbers

-
Team assignments

-
Individual participant selection

-
Session configuration

-
Match configuration

03. Game Library

Show exactly three games:

-
Football AI

-
Falaj Flow

-
Souq Signal

Each must have:

-
Original illustrations

-
Game description

-
Rules

-
Team/individual label

-
Estimated duration

-
Equipment requirements

-
Start button

04. Game Setup

Show:

-
Player assignment

-
Team setup

-
Equipment checklist

-
Camera calibration

-
Video/audio permission checks

-
Scoring rules

-
Start countdown

05. Live Game Arena

Include:

-
Camera preview

-
Upload/record controls

-
Active players

-
Timer

-
Round number

-
Match score

-
Processing status

-
Observed events

06. AI Referee Replay

Show:

-
Actual gameplay footage

-
Annotated evidence

-
Relevant timestamps

-
AI observations

-
Rule references

-
Score changes

-
Verification status

-
Confidence/uncertainty indicators

-
Human-review controls

07. Family Cup Results

Include:

-
Individual standings

-
Team standings

-
Game winners

-
MVP where justified

-
Score breakdowns

-
Championship history

-
Export results

08. AI Evaluation Lab

Include:

-
Test scenario selection

-
Expected results

-
Actual results

-
Rule-validation status

-
Detection limitations

-
Measured processing duration

-
Repeatable evaluation runs

All screens must be functional.

6. Design direction

Develop a premium editorial interface.

Color palette:

-
Midnight navy #12243A

-
Warm ivory #F7F3EB

-
Heritage teal #167E80

-
Emirates red #B92D39

-
Muted gold #C6A36B

-
Charcoal #26313C

Use:

-

Large cinematic typography

-
Beautiful game cards

-
Subtle motion

-
Bespoke AI visualizations

-
High-quality icons

-
Structured navigation

-
Refined loading animations

-
Responsive layouts

-
Accessible contrast

-
Elegant score animations

Avoid:

-
Generic AI chatbot styling

-
Excessive neon

-
Cluttered dashboards

-
Fake technical visualizations

-
Nonfunctional buttons

-
Placeholder pages

The result should resemble a premium, professionally engineered product.

PART V — TECHNICAL ARCHITECTURE

7. Technology

Frontend

- React
- TypeScript
- Vite
- Tailwind CSS
- Canvas/SVG visualizations
- MediaRecorder API
- Responsive application routing

Backend

- Python
- FastAPI
- OpenCV
- FFmpeg
- Object detection/tracking where feasible
- Speech transcription
- Pydantic validation
- Deterministic scoring rules

Optional AI integration

- Gemini multimodal API for structured interpretation and explanation

All API credentials must remain server-side.

The final scoring authority must be the validated rules engine, not an unconstrained language-model response.

8. AI processing workflow

Implement:

-
Receive video/audio.

-
Validate media format and duration.

-
Calculate source SHA-256 hash.

-
Extract timestamped frames and audio.

-
Run game-specific perception.

-
Produce structured observations.

-
Validate observations.

-
Apply deterministic game rules.

-
Generate evidence-backed event reports.

-
Flag uncertain events.

-
Present results for host review.

-
Apply verified scoring events.

-
Update championship standings.

Build three distinct game-specific judging adapters.

Use actual model outputs and computer vision observations.

Do not hardcode successful results into the application.

9. Backend endpoints

Implement appropriately designed endpoints for:

- Session creation
- Player management
- Game configuration
- Media upload
- Game analysis
- Judging reports
- Result verification
- Score application
- Match history
- Evaluation execution
- Health checks

Provide explicit request and response schemas.

10. Accuracy and fairness

Implement:

- Rule-version tracking
- Source-video hashes
- Duplicate-event prevention
- Input validation

-
Human override audit trail

-
Uncertain-event handling

-
Score integrity verification

-
Error logging

Do not display invented benchmark results or accuracy percentages.

If an AI event is uncertain, explain the limitation.

Clearly distinguish observed facts, derived estimates and human decisions.

PART VI — GITHUB AND VERCEL DEPLOYMENT

11. Repository

Create a clean, standalone GitHub-ready project.

Suggested repository name:

wasl-arena

Do not modify or overwrite the existing DOMINO repository.

Use an organized structure such as:

-
/frontend

-
/backend

-
/public

-
/assets

-
/tests

-
/docs

-
/presentation

-
/video

-
/ai-usage

Include:

-
README

-
Requirements and setup documentation

-
package.json

-
requirements.txt

-
.env.example

-
.gitignore

-
Vercel configuration

-
Backend deployment configuration

-
Automated tests

12. Deployment

The frontend will be imported from GitHub into Vercel.

Prepare it for this deployment.

Requirements:

-
Production build succeeds.

-
SPA routes work after refresh.

-
Environment variables are configurable.

-
Static assets resolve correctly.

-
No secrets are exposed in browser code.

-
Frontend communicates with a separately hosted Python backend.

Use Render or another suitable persistent Python host for video processing.

Do not assume Vercel frontend hosting alone is sufficient for computationally intensive video analysis.

Document all required deployment settings.

Do not claim deployment success without verifying the live application.

PART VII — TECHNICAL REPORT

13. Produce a six-page technical report

After the working prototype is developed and tested, generate the complete technical report.

Length: Exactly six pages.

Deliverables:

-
Editable Word document

-
Final PDF

Design:

-
Premium academic formatting

-
Navy, ivory and teal accents

-
Consistent headings

-
Technical diagrams

-
Well-formatted tables

-
Genuine prototype screenshots

-
Readable references

Required structure

Page 1 — Executive Summary and Problem Definition

-
Challenge background

-
Family gaming problem

-
Project objectives

-
Proposed solution

-
Novelty and intended impact

Page 2 — Games and User Experience

-
Three-game overview

-
Family registration workflow

-
Team and individual gameplay

-
Host controls

-
Rules and scoring

-
User journey

Page 3 — Technical Architecture

-
System architecture diagram

-
Frontend/backend components

-
Video and audio processing

-
AI pipeline

-

Rule validation

-

API integration

Page 4 — AI Methodology

-

Football detection and tracking

-

Falaj graph reconstruction

-

Souq multimodal verification

-

Evidence extraction

-

Structured outputs

-

Explainability

Page 5 — Testing and Evaluation

-

Evaluation methodology

-

Test scenarios

-

Actual test outcomes

-

Scoring correctness

-

Failure cases

-

Limitations

Page 6 — Impact, Deployment and Conclusions

-

Family engagement

-

Cultural relevance

-

Fairness and privacy

-
Deployment architecture

-
Key achievements

-
Future improvements

-
References

Use genuine measurements and verified implementation details.

Do not fabricate test results, deployment URLs, citations or performance metrics.

PART VIII — PRESENTATION

14. Generate exactly 30 slides

Produce a complete, professional PowerPoint presentation.

Format: Editable .pptx.

Visual quality: High-end competition pitch deck.

Follow the WASL Arena design system.

Include authentic screenshots, technical diagrams, scoring examples and evaluation evidence.

Exact slide sequence

-
WASL Arena — Title

-
One Family. Three Games. One Fair AI Referee.

-
The Problem

-
Why Family Gaming Matters

-
ADU Challenge Requirements

-
Our Solution

-
The WASL Arena Experience

-
What Makes WASL Arena Different

-
User Journey

-
Three-Game Collection

-
Football AI — Overview

-
Football AI — Detection Architecture

-
Football AI — Referee Evidence

-
Falaj Flow — Overview

-
Falaj Flow — Graph Intelligence

-
Falaj Flow — AI Scoring Demonstration

-
Souq Signal — Overview

-
Souq Signal — Multimodal AI

-
Souq Signal — Verified Transaction

-
Human Host Console

-
AI Referee Replay

-
End-to-End System Architecture

-
AI Models and Algorithms

-
Scoring Integrity and Explainability

-
Fairness, Privacy and Safety

-
Testing and Evaluation Methodology

-
Results and Demonstrated Capabilities

-
GitHub and Vercel Deployment

-
Limitations and Future Development

-
Conclusion — Play Together. Trust the Referee.

Slide requirements

Every slide must:

-
Have a clear central message

-
Use concise presentation-level wording

-
Include relevant visuals

-
Avoid overcrowding

-
Connect to the competition requirements

Use actual prototype screenshots for the game and UI demonstrations.

Show the technical sophistication of the solution without unnecessary jargon.

Do not invent statistics.

Include speaker notes where appropriate.

PART IX — TWO-MINUTE VIDEO

15. Produce a complete cinematic video production package

Create a competition-ready video demonstrating WASL Arena.

Maximum duration: 120 seconds.

Format: 16:9, 1920×1080, MP4.

Scene structure: Exactly 24 clips, each lasting five seconds.

Total runtime: 24 × 5 = 120 seconds.

Do not add separate title cards, transitions or outros that extend the runtime.

All transitions must fit within the allotted clip durations.

Required video outputs

For each of the 24 clips, generate:

-
Clip number

-
Start and end timestamp

-
Scene purpose

-
Detailed visual description

-
Exact five-second AI-video-generation prompt

-
Camera framing and movement

-
Required screenshot or screen recording

-
Narration text

-
On-screen text

-
Transition direction

-
Music/SFX guidance

The clips must form one cohesive story.

Storyboard structure

ACT 1 — THE PROBLEM AND VISION

-
Clip 01: Cinematic opening — family gaming

-

Clip 02: The difficulty of tracking gameplay fairly

-
Clip 03: Introducing WASL Arena

-
Clip 04: Three games, one AI referee

ACT 2 — THE APPLICATION

-
Clip 05: App landing-page screenshot/recording

-
Clip 06: Player and team registration

-
Clip 07: Three-game library

-
Clip 08: Host controls and match setup

ACT 3 — FOOTBALL AI

-
Clip 09: Real mini-football gameplay

-
Clip 10: Football video being processed

-
Clip 11: Player and ball tracking

-
Clip 12: Verified goal and score update

ACT 4 — FALAJ FLOW

-
Clip 13: Physical Falaj board gameplay

-
Clip 14: AI identifies channel tiles

-
Clip 15: Digital network reconstruction

-
Clip 16: Verified water delivery and scoring

ACT 5 — SOUQ SIGNAL

-
Clip 17: Physical marketplace challenge

-
Clip 18: Spoken and visual item recognition

-
Clip 19: AI verifies the budget and basket

-
Clip 20: Verified shopping result

ACT 6 — PROOF AND CONCLUSION

-
Clip 21: AI Referee Replay evidence

-
Clip 22: Evaluation and technical proof

-
Clip 23: Family Cup championship results

-
Clip 24: WASL Arena closing statement

Screenshot requirements

Capture actual application screenshots only after the frontend is implemented.

Include genuine screen recordings or screenshots of:

-
Landing page

-
Family registration

-
Game selection

-
Football results

-
Falaj network visualization

-
Souq recognition

-
AI Referee Replay

-
Final leaderboard

Do not generate fictional UI screenshots and present them as the implemented product.

Use authentic gameplay footage to demonstrate AI detections and scoring.

Narration

Write concise, natural, engaging narration.

The script must be:

- Technically accurate
- Easy for judges to understand
- Confident
- Memorable
- Competition appropriate

Each five-second segment must contain narration that can comfortably be spoken within its duration.

Use a single consistent narration style throughout.

Music and style

- Premium cinematic atmosphere
- Modern technological sound
- Subtle UAE heritage references
- Clear voice-over
- Tasteful kinetic typography
- Smooth transitions
- Readable captions

Avoid excessive effects and distracting visual clutter.

If video generation or editing tools are unavailable, produce the complete scene-by-scene script, generation prompts, editing timeline, asset manifest and export instructions.

PART X — AI CHAT USAGE REPORT

16. Generate the AI-development usage report

The competition requires all AI chats to be submitted as a single PDF.

Produce a comprehensive, transparent documentation package covering AI assistance during development.

Include

- Project name
- Team identification fields
- AI tools used
- Models used when known
- Purpose of each tool
- Development workflow
- Prompting methodology
- Architecture discussions
- Code-generation interactions
- Debugging interactions
- Design iterations
- Technical report assistance
- Presentation assistance
- Video scripting assistance

-
Human review and verification

Most important requirement

Preserve the original AI conversation records.

Do not invent conversations, model outputs, timestamps, tests or development activities.

Where original chat exports are available, compile the authentic transcripts into one PDF.

Where chats have not yet been supplied, prepare clearly labeled placeholders and an import workflow.

Redact credentials, API keys, tokens and sensitive personal information before creating the final submission PDF.

Explain which outputs were verified or modified by the team.

The final report must be transparent, authentic and professionally organized.

PART XI — VERIFICATION AND QUALITY ASSURANCE

17. Required testing

Create an executable evaluation suite.

Football

-
Valid goal

-
No-goal scenario

-
Uncertain ball detection

-
Score calculation

-
Tie-break behavior

Falaj Flow

-
Fully connected irrigation network

-
Partially connected network

-
Blocked route

-
Invalid tile orientation

-
Correct score calculation

Souq Signal

-
Correct item selection

-
Wrong item selection

-
Speech/video disagreement

-
Exceeded budget

-
Duplicate selection

-
Uncertain audio

Cross-platform

-
Camera permissions

-
File upload

-
API failures

-
Session persistence

-
Result validation

-
Mobile responsiveness

-
Deployment routing

Report actual test outcomes.

Do not mark tests as passed unless they execute successfully.

PART XII — FINAL DELIVERY

18. Required submission items

Prepare exactly the following five principal deliverables.

1. Working Prototype

-
Complete source code

-
GitHub repository

-
Vercel-deployed frontend

-
Working processing backend

-
Functional three-game AI judging

2. Technical Report

-
Six-page report

-
Editable Word version

-
Final PDF

3. AI Chat Usage Report

-
Complete authentic compiled AI interaction PDF

-
Credentials redacted

4. Presentation

-
Thirty-slide PowerPoint

-
Actual prototype screenshots

-
Technical and evaluation content

5. Video

-
Maximum 120-second video

-
Twenty-four five-second clips

-
Complete narration

-
Detailed generation prompts

-
Real app screenshots and gameplay demonstrations

19. Development workflow

Follow this order:

PHASE 1 — Architecture
Define the data structures, game rules, judging algorithms, backend contract and deployment strategy.

PHASE 2 — Prototype Development
Build the complete frontend, backend, AI adapters and scoring system.

PHASE 3 — Technical Verification
Execute tests, fix failures, collect real gameplay evidence and verify judging outputs.

PHASE 4 — Deployment
Prepare the GitHub repository, deploy or provide exact deployment instructions, connect Vercel to the frontend and configure the backend.

PHASE 5 — Documentation
Generate the six-page technical report using actual implementation evidence.

PHASE 6 — Presentation
Create the thirty-slide deck using the validated architecture, app screenshots and real results.

PHASE 7 — Video Production
Generate the complete twenty-four-clip storyboard, individual video-generation prompts, narration, captions and editing instructions.

PHASE 8 — AI Usage Documentation
Compile and redact authentic AI conversations into one submission-ready PDF.

PHASE 9 — Final Audit
Check every submission against the official ADU requirements.

20. Critical instructions

-
The track is Cycle 3.

-
Exactly three games.

-
Football is mandatory.

-
The other two games must be original.

-
Both team and individual modes are required.

-
AI must analyze real video/audio evidence.

-
Every important score must be explainable.

-
No invented AI outcomes.

-
No fictional performance metrics.

-
No hardcoded judging demonstrations presented as live analysis.

-
No exposed API credentials.

-
No unverified deployment claims.

-
No fabricated AI conversations.

-
No additional games beyond the three specified.

-
Do not overwrite DOMINO.

-
Prioritize functioning engineering over decorative complexity.

21. What you must do now

Begin with a concise, concrete architecture and implementation specification for WASL Arena.

Then proceed directly to development.

Do not stop after brainstorming or creating a plan.

Implement and validate the application first, then use the actual implementation to generate the technical report, presentation and video materials.

At every stage, distinguish completed work from pending work.

If any external action requires authorization, such as publishing to GitHub, deploying a service or supplying API credentials, identify it explicitly and continue all independent work.

The final product must be technically credible, visually outstanding and unmistakably aligned with the Cycle 3 Family AI Game Day competition.

Our objective is to deliver the strongest possible competition entry through originality, functional intelligence, transparent judging, exceptional user experience and verifiable technical evidence.

My final strategic recommendations

1. Prioritize the AI referee over visual complexity.

A real five-second clip producing a correct, explainable judgment is more persuasive than a polished interface with simulated results.

2. Record genuine gameplay early.

Football, Falaj Flow and Souq Signal each need at least one usable recording. These recordings will support the prototype, tests, report, presentation and video simultaneously.

3. Build the five submission items from the same verified evidence.

Use the same screenshots, architecture, test results and AI-referee decisions across the technical report, thirty-slide presentation and two-minute demonstration.

My verdict: WASL Arena with Football AI, Falaj Flow and Souq Signal is the strongest direction I recommend for your updated Cycle 3 challenge. It combines sports, Emirati heritage, visual intelligence, multimodal AI and explainable scoring within exactly three games. Its winning potential will depend on whether the actual video-based judging works reliably during the demonstration.

USER · message 3

We are running low on time, integrate all prompts in one, and also include links of designs from spline and recent.

CHATGPT · message 4

May, here is the single consolidated master prompt. It covers the entire competition submission, including development, deployment, all documentation, the 30-slide presentation, and the 24 five-second video clips.

I've also incorporated verified references from Spline and Recent.design, including specific interactive scenes and design examples. Spline supports embedding interactive 3D in React applications, while Recent provides references for interface transitions and visual storytelling.

spline.design+3

Ready to paste into Claude

ULTIMATE MASTER EXECUTION PROMPT — WASL ARENA

ULTIMATE MASTER EXECUTION PROMPT — WASL ARENA

6th Annual Abu Dhabi University COE STEM Programming with AI Competition

Act as an elite team of full-stack engineers, AI researchers, computer vision specialists, creative directors, technical writers, presentation designers and video production experts.

Your mission is to develop WASL Arena and produce its complete competition submission in one coordinated workflow.

Do not stop at brainstorming, planning, wireframes or pseudocode. Implement the application, validate its functionality and generate the required submission deliverables.

I am working under a strict deadline. Make reasonable technical and design decisions autonomously, minimize unnecessary clarification questions, and prioritize functional results.

Use all attached ADU competition documents as the authoritative requirements.

1. COMPETITION REQUIREMENTS

Competition: 6th Annual COE STEM Programming with AI Competition

Theme: AI for Family

Challenge: Family AI Game Day

Category: Cycle 3 — Grades 9–12

The official problem requires a functioning website or application in which families play games inspired by sports, culture or heritage.

A human host manages the competition, while AI observes actual gameplay through video and audio and applies the game rules fairly.

Mandatory requirements

-
Exactly three games: football plus two original games.

-
Both team and individual competition.

-
Player and team registration.

-
Rules, scoring, rounds, timers and penalties.

-
Tie-breaks and winning conditions.

-
Uploading gameplay video with audio.

-
Direct camera and microphone recording.

-
AI detection of actual gameplay events.

-
Explainable AI judgments.

-
Verified score updates.

-
Final results and MVP where applicable.

-
Functional end-to-end demonstration.

Do not develop five games or use the example games prohibited in the official question.

Required submission items

Produce:

-
Working interactive prototype.

-
Six-page technical report.

-
Thirty-slide PowerPoint presentation.

-
AI chat usage report containing authentic development conversations.

-
Video advertisement/demonstration of no more than two minutes.

The application must be published through GitHub and imported into Vercel.

2. PRODUCT CONCEPT

WASL Arena — The Family AI Games Championship

Tagline: One family. Three games. One fair AI referee.

Develop an original, premium AI-powered gaming platform combining:

-
Real physical gameplay

-
Computer vision

-
Multimodal AI

-
Audio recognition

-
Deterministic scoring

-

Transparent judgment explanations

-
UAE heritage

-
Family interaction

The most important innovation is:

AI REFEREE REPLAY

Every scoring event must be backed by identifiable evidence.

For each judgment, show:

-
Source video

-
Relevant timestamps

-
Detected events

-
Visual annotations

-
Audio evidence when applicable

-
Applied rule

-
Score change

-
Verification status

-
Uncertainty indication

-
Human-review option

For example, a football goal must be linked to an observed ball crossing the defined goal boundary.

Do not fabricate goals, detections, transcripts, scores or confidence values.

The judges must be able to understand why the AI reached a decision.

3. THE THREE GAMES

GAME 1: FOOTBALL AI

Mode: Team versus team.

Implement:

-
Player registration

-
Team colors and numbers

-
Ball/player tracking

-
Goal-area calibration

-
Goal detection

-
Match timer

-
Score calculation

-
Conservative possession estimates

-
Winner determination

-
MVP based on verified contributions

-
AI Referee Replay

Use OpenCV and appropriate object detection/tracking.

A controlled mini-football setup with a fixed camera, bright ball, clearly marked goals and contrasting shirts is acceptable for the demonstrable prototype, provided the recorded activity is genuine.

Rules

-
Valid goal: +1

-
Invalid or unsupported goal: 0

-
Winner: Highest verified score

-
Tie: Configurable golden-goal extension

-
MVP: Verified individual contribution

Never present uncertain tracking as a verified observation.

GAME 2: FALAJ FLOW

Mode: Team versus team.

Create a physical strategy challenge inspired by traditional UAE irrigation.

Players construct a water-channel network using printed tiles on a 5×5 board.

The AI identifies the tile positions and orientations, reconstructs the network as a graph and verifies whether water can reach three agricultural destinations.

AI capabilities

-
Visual board recognition

-
Tile-position detection

-
Graph construction

-
Route validation

-
Obstacle detection

-
Automated scoring

-
Animated, evidence-backed water-flow visualization

Rules

-
Connected destination: +3

-
Invalid outlet/leak: -1

-
Time bonus: Up to +2

-
Round duration: 90 seconds

-
Tie-break: Shortest valid network, then time

Wow factor: Split-screen physical board and live digital reconstruction, with verified water flowing across the graph.

GAME 3: SOUQ SIGNAL

Mode: Individual competition.

Create an AI-observed shopping challenge inspired by UAE marketplaces.

Players receive a shopping mission and select physical product cards under a limited budget.

The cards represent products such as dates, coffee, spices, pottery and textiles.

Each card has:

-
Unique visual marker

-
Fixed price

-
Product name

-
Arabic/English labels

The player displays selected cards to the camera and verbally announces the products.

AI capabilities

-
Visual item recognition

-
Speech transcription

-
Audio/video agreement checking

-
Budget calculations

-
Duplicate detection

-
Rule validation

-
Automated scoring

Rules

-
Correct requested item: +2

-
Valid matching spoken selection: +1

-
Exceeded budget: -2

-
Duplicate item: No additional item points

-
Tie-break: Lower valid expenditure, then time

Wow factor: Live visual recognition, synchronized speech transcript, verified basket, remaining budget and explainable AI referee.

4. PREMIUM DESIGN SYSTEM

The application must look like a commercially polished, award-worthy product.

Visual identity

Colors:

-
Midnight navy: #12243A

-
Ivory: #F7F3EB

-
Heritage teal: #167E80

-
Emirates red: #B92D39

-
Muted gold: #C6A36B

-
Charcoal: #26313C

Use:

-
Modern editorial typography

-

Elegant spacing

-
Strong typographic hierarchy

-
Cinematic hero sections

-
Interactive game cards

-
Refined motion and microinteractions

-
Premium scoreboards

-
Responsive layouts

-
Authentic UAE cultural references

Avoid generic templates, neon aesthetics, excessive glassmorphism, unreadable effects and unnecessary decorative complexity.

REQUIRED DESIGN REFERENCES

Research the following references before implementing the interface.

SPLINE — Interactive 3D

Spline for Web:
https://spline.design/web

Use this reference to understand how interactive 3D scenes integrate into React.

Reactive Orb — Spline Community:
https://app.spline.design/community/file/306ca2a5-d1ab-46ac-a27c-198575c82db0

Adapt its reactive-motion concept for WASL Arena's AI Referee visualization.

Create an original visual that reacts to analysis states:

-
Idle

-
Watching

-
Processing

-
Evidence verified

-
Review required

Reeded Glass / Prism Hero Concept:
https://app.spline.design/community/file/a74713b7-6154-4b89-a5d9-47cfca3f413f

Study subtle depth, lighting and premium composition. Do not replicate its entire visual style.

Spline Community:
https://community.spline.design/

Explore suitable interactive elements for game selection and hero animations.

RECENT.DESIGN — Contemporary Interaction Design

Website Inspiration:
https://recent.design/websites

Study premium product layouts, transitions, navigation, composition and typography.

Krea Image Generation Loading State:
https://recent.design/i/f3cwpo5-ai-image-generation-loading-state

Reference the progressive reveal of AI-processing stages.

Apply the underlying interaction principle to WASL Arena's judging sequence:
Uploading → Extracting Evidence → Detecting Events → Applying Rules → Verified Result.

Krea Agent Onboarding:
https://recent.design/i/t865vsj-krea-agent-onboarding

Reference its structured onboarding and transitions for family registration, team setup and game configuration.

Linearity AI Launch Video:
https://recent.design/i/3qpnwhw-linearity-ai-launch

Study concise product storytelling, interface-focused motion and clean transitions for the competition video.

Design integration requirements

Do not copy complete interfaces, branding or protected design assets.

Use the references to inform original layouts, interactions and 3D elements.

Only embed a Spline scene if it is permitted for reuse and technically reliable.

If a specific scene is inaccessible or unsuitable, recreate the design principle using original React, CSS, SVG or Three.js implementations.

Optimize performance and provide a lightweight fallback if interactive 3D cannot load.

Do not let visual enhancements interfere with gameplay or judging reliability.

5. APPLICATION SCREENS

Develop eight complete screens:

-
Launch: Cinematic branding, interactive hero, championship introduction.

-
Family Setup: Players, teams, language, match configuration.

-
Game Library: Exactly three visually distinctive game cards.

-
Game Preparation: Instructions, equipment, rules, calibration.

-
Live Arena: Camera, recording, timer, rounds, scores.

-
AI Referee Replay: Evidence, detections, rules and verification.

-
Family Cup: Results, winners, MVP and standings.

-
Evaluation Lab: Actual test cases, outputs and limitations.

Every button, link and navigation action must work.

Provide downloadable printable game boards and cards for Falaj Flow and Souq Signal.

6. TECHNICAL IMPLEMENTATION

Frontend

-
React

-
TypeScript

-
Vite

-
Tailwind CSS

-
Responsive routing

-
MediaRecorder API

-

Canvas/SVG annotations

-
Spline integration where useful

Backend

-
Python

-
FastAPI

-
OpenCV

-
FFmpeg

-
Appropriate pretrained vision models

-
Speech transcription

-
Pydantic schemas

-
Game-specific deterministic rule engines

Gemini may support structured interpretation, transcription or explanations, but it must not invent events or serve as an unchecked scoring authority.

Processing pipeline

Gameplay → Video/Audio → Media Extraction → Visual/Audio Detection → Structured Events → Rule Validation → Evidence Report → Host Review → Verified Score.

Implement real processing for all three games.

Every major event must reference its underlying evidence.

Create:

-
Source-video hashes

-
Rule-version identifiers

-
Event timestamps

-
Structured judging reports

-
Validation results

-
Review states

-
Session audit history

Uncertain detections must be labeled NEEDS_REVIEW.

Never hardcode fake positive results into the live judging workflow.

Testing

Include executable tests for:

Football

-
Valid goal

-
No-goal

-
Uncertain detection

-
Score update

-
Tie-break

Falaj Flow

-
Complete valid route

-
Partial network

-
Blocked route

-
Invalid connection

-
Correct scoring

Souq Signal

-
Correct selection

-
Incorrect selection

-
Audio/video mismatch

-
Budget violation

-
Duplicate selection

Report only tests that actually ran.

Use authentic recorded gameplay for at least one end-to-end demonstration per game.

7. GITHUB AND VERCEL

Create a standalone repository named:

wasl-arena

Do not modify DOMINO.

Organize frontend, backend, tests, documentation, presentation assets, video assets and AI-usage records.

Prepare:

-
README.md

-
package.json

-
requirements.txt

-
.env.example

-
.gitignore

-
vercel.json where needed

-
Deployment configuration

-
Backend setup guide

-
Test instructions

The React frontend must deploy through GitHub → Vercel.

Use Render or another appropriate backend host for video processing.

Configure the backend URL through environment variables.

Keep all credentials server-side.

Ensure:

-
Successful production build

-
Functional routes

-
Responsive design

-
Working API connection

-
No missing assets

-
No exposed secrets

Do not claim that publication is complete until the live application is verified.

8. SIX-PAGE TECHNICAL REPORT

Generate the complete six-page technical report.

Output:

-
Editable Word document

-
Final PDF

Use a professional academic design, with consistent navy/teal accents and real application screenshots.

Page 1: Executive Summary, Problem and Objectives

Page 2: System Concept, Three Games and User Journey

Page 3: System Architecture, Technology and AI Pipeline

Page 4: Football, Falaj and Souq AI Methodologies

Page 5: Testing, Results, Validation and Limitations

Page 6: Fairness, Privacy, Deployment, Impact and Conclusions

Include diagrams, tables, references and only genuine technical results.

Exactly six pages.

9. THIRTY-SLIDE PRESENTATION

Generate an editable PowerPoint with exactly 30 slides.

Use the WASL Arena design language and genuine screenshots.

Slide sequence:

-
Title

-
Vision

-
Problem

-
Why Family Gaming Matters

-
Official Challenge

-
Our Solution

-
Value Proposition

-
Signature Innovation

-
User Journey

-
Three-Game Collection

-
Football Overview

-

Football AI Tracking

-

Football Referee Replay

-

Falaj Flow Overview

-

Falaj Graph Intelligence

-

Falaj Scoring

-

Souq Signal Overview

-

Multimodal AI

-

Souq Transaction Verification

-

Family Host Console

-

AI Referee Replay

-

Technical Architecture

-

AI Models and Algorithms

-

Evidence Integrity

-

Fairness and Privacy

-

Evaluation Methodology

-

Actual Test Results

-

GitHub and Vercel Deployment

-

Limitations and Future Development

-

Conclusion and Final Impact

Make every slide visually cohesive, concise and technically meaningful.

Include presenter notes.

Do not fabricate evaluation data.

10. TWO-MINUTE VIDEO — 24 CLIPS

Develop a complete cinematic video production package.

Maximum length: 120 seconds.

Clip length: Exactly five seconds per clip.

Total: 24 clips × 5 seconds.

Format: 16:9, 1080p MP4.

Transitions must be included within the clip durations. Do not add extra runtime.

Required storyboard

ClipContent

01Cinematic family-gaming opening

02The problem of fair game judging

03WASL Arena brand reveal

04Three games, one AI referee

05Actual app landing page

06Family registration

07Three-game library

08Host console

09Real football gameplay

10Football video analysis

11Player and ball tracking

12Goal verification and score

13Falaj physical gameplay

14Visual tile recognition

15Digital irrigation graph

16Verified Falaj scoring

17Souq physical gameplay

18Visual and speech recognition

19Budget and item verification

20Souq results

21AI Referee Replay

22Testing and technical evidence

23Family Cup leaderboard

24Cinematic closing statement

For each clip, produce:

-
Exact five-second scene description.

-
Detailed AI-video-generation prompt.

-
Camera framing and movement.

-
Source asset or real app screenshot.

-
Exact narration.

-
On-screen text.

-
Sound effects/music guidance.

-
Transition to next clip.

Mandatory video rules

-
Use actual screenshots captured from the completed application.

-
Use genuine gameplay footage for scoring demonstrations.

-
Do not fabricate functioning UI or AI evidence.

-
Keep narration short enough for five seconds.

-
Use consistent colors, typography and atmosphere.

-
Make the video emotionally engaging and technically credible.

-
Generate a full narration transcript.

-
Generate a shot-by-shot editing timeline.

-
Provide an asset checklist.

-
Provide final export settings.

Use the Recent.design launch-video reference to guide the visual pacing, but create an original film.

If direct MP4 production is unavailable, generate a complete production-ready storyboard, prompts, narration, timing sheet and editing instructions.

11. AI CHAT USAGE REPORT

Prepare a single PDF documenting authentic AI assistance throughout development.

Include:

-
AI tools and models

-
Development phases

-
Architecture prompts

-
Code generation

-
Debugging

-
Testing

-
Design iterations

-
Documentation creation

-
Presentation development

-
Video scripting

-
Human verification and adjustments

Import original conversation exports where provided.

Do not invent chat transcripts.

If exports are unavailable, identify them as pending and provide a clear compilation process.

Redact API credentials and sensitive information.

The report must accurately represent the team's actual AI usage.

12. EXECUTION PRIORITIES

Follow this strict order.

Priority 1 — Prototype
Build all three games, real judging adapters, host controls and evidence replay.

Priority 2 — Technical verification
Test the application and collect genuine screenshots, recordings and results.

Priority 3 — GitHub/Vercel
Prepare and verify the production deployment.

Priority 4 — Technical report
Generate the six-page document from actual implementation evidence.

Priority 5 — Presentation
Generate thirty editable slides using genuine prototype results.

Priority 6 — Video
Produce the twenty-four five-second clip prompts, narration, screenshots and editing package.

Priority 7 — AI chat report
Compile authentic AI conversations into one PDF.

Priority 8 — Final submission audit
Verify every deliverable against the official brief.

Work autonomously and minimize interruptions.

If external authorization or credentials are required, identify the blocker and continue everything that can be completed safely.

13. NON-NEGOTIABLE FINAL QUALITY CHECKS

Before declaring completion, verify:

-
Exactly three games.

-
Football included and functional.

-
Two original games implemented.

-
Both team and individual play.

-
Real video/audio-based AI judging.

-
Evidence-backed scoring.

-
Functional host controls.

-
Complete navigation.

-
Actual test results.

-
No fabricated demonstrations.

-
No exposed credentials.

-
Working production build.

-
GitHub/Vercel deployment prepared and verified where access permits.

-
Six-page technical report.

-
Thirty-slide presentation.

-
Authentic AI chat PDF.

-

Twenty-four five-second video clips or complete production package.

-

Total video runtime no more than 120 seconds.

Begin implementation immediately. Build the prototype first, then generate every submission artifact from the verified application.

FINAL OBJECTIVE: Deliver a distinctive, polished, fully functional and technically defensible Cycle 3 competition entry, demonstrating that AI can make family gaming more interactive, inclusive, culturally meaningful and transparently fair.

One important priority given the deadline: Have Claude complete the working prototype and capture its screenshots before producing the report, slides or video. Those three deliverables can then share the same real evidence, saving substantial time while keeping the submission consistent.

USER · message 5

paste the spline and recent links here in the chat

CHATGPT · message 6

Spline — Interactive 3D Design

-

Spline for Web — Interactive 3D Integration

-

Reactive 3D Orb — AI Referee Animation

-

Reeded Glass / Prism Hero Concept

-

Spline Community — 3D Templates and Interactions

Recent.design — UI/UX and Motion Inspiration

-

Recent.design — Website Inspiration

-

Krea AI — Animated Loading States

-

Krea Agent — Interactive Onboarding

-

Linearity AI — Cinematic Launch Video

My top three for WASL Arena:

-

Spline Reactive Orb: For the AI Referee's interactive visualization.

-

Krea AI Loading States: For video analysis, event detection and score verification.

-

Recent.design Website Gallery: For the landing page, game-selection cards and overall interface design.

If you want, I can:

-

Create detailed implementation plan for integrating Spline and Recent.design links

-

Draft AI Referee visualization concept based on Spline Reactive Orb link

-

Outline UI motion sequences inspired by Recent.design loading and onboarding states

Appendix B: Claude Code conversation (verbatim user messages and assistant replies)

Tool calls and tool outputs are omitted for length (summarised in section 5). Pasted API credentials are redacted.

USER · 2026-10-29 11:20:56 UTC

@"C:\Users\aaa\Desktop\6th Annual ADU CoE STEM Hackathon\Competition Documents\6th Annual STEM ADU COE Hackathon.pdf"
@"C:\Users\aaa\Desktop\6th Annual ADU CoE STEM Hackathon\Competition Documents\COE Competitions and Successes.pdf"

ULTIMATE MASTER EXECUTION PROMPT — WASL ARENA
6th Annual Abu Dhabi University COE STEM Programming with AI Competition

Act as an elite team of full-stack engineers, AI researchers, computer vision specialists, creative directors, technical writers, presentation designers and video production experts.

Your mission is to develop WASL Arena and produce its complete competition submission in one coordinated workflow.

Do not stop at brainstorming, planning, wireframes or pseudocode. Implement the application, validate its functionality and generate the required submission deliverables.

I am working under a strict deadline. Make reasonable technical and design decisions autonomously, minimize unnecessary clarification questions, and prioritize functional results.

Use all attached ADU competition documents as the authoritative requirements.

---

1. COMPETITION REQUIREMENTS

Competition: 6th Annual COE STEM Programming with AI Competition

Theme: AI for Family

Challenge: Family AI Game Day

Category: Cycle 3 — Grades 9–12

The official problem requires a functioning website or application in which families play games inspired by sports, culture or heritage.

A human host manages the competition, while AI observes actual gameplay through video and audio and applies the game rules fairly.

Mandatory requirements

- Exactly **three games: football plus two original games**.
- Both team and individual competition.
- Player and team registration.
- Rules, scoring, rounds, timers and penalties.
- Tie-breaks and winning conditions.
- Uploading gameplay video with audio.
- Direct camera and microphone recording.
- AI detection of actual gameplay events.
- Explainable AI judgments.
- Verified score updates.
- Final results and MVP where applicable.

- Functional end-to-end demonstration.

Do not develop five games or use the example games prohibited in the official question.

Required submission items

Produce:

1. Working interactive prototype.
2. Six-page technical report.
3. Thirty-slide PowerPoint presentation.
4. AI chat usage report containing authentic development conversations.
5. Video advertisement/demonstration of no more than two minutes.

The application must be published through GitHub and imported into Vercel.

---

2. PRODUCT CONCEPT

WASL Arena — The Family AI Games Championship

Tagline: One family. Three games. One fair AI referee.

Develop an original, premium AI-powered gaming platform combining:

- Real physical gameplay
- Computer vision
- Multimodal AI
- Audio recognition
- Deterministic scoring
- Transparent judgment explanations
- UAE heritage
- Family interaction

The most important innovation is:

AI REFEREE REPLAY

Every scoring event must be backed by identifiable evidence.

For each judgment, show:

- Source video
- Relevant timestamps
- Detected events
- Visual annotations
- Audio evidence when applicable
- Applied rule
- Score change
- Verification status
- Uncertainty indication
- Human-review option

For example, a football goal must be linked to an observed ball crossing the defined goal boundary.

Do not fabricate goals, detections, transcripts, scores or confidence values.

The judges must be able to understand why the AI reached a decision.

---

3. THE THREE GAMES

GAME 1: FOOTBALL AI

Mode: Team versus team.

Implement:

- Player registration
- Team colors and numbers
- Ball/player tracking
- Goal-area calibration
- Goal detection
- Match timer
- Score calculation
- Conservative possession estimates
- Winner determination
- MVP based on verified contributions
- AI Referee Replay

Use OpenCV and appropriate object detection/tracking.

A controlled mini-football setup with a fixed camera, bright ball, clearly marked goals and contrasting shirts is acceptable for the demonstrable prototype, provided the recorded activity is genuine.

Rules
- Valid goal: +1
- Invalid or unsupported goal: 0
- Winner: Highest verified score
- Tie: Configurable golden-goal extension
- MVP: Verified individual contribution

Never present uncertain tracking as a verified observation.

GAME 2: FALAJ FLOW

Mode: Team versus team.

Create a physical strategy challenge inspired by traditional UAE irrigation.

Players construct a water-channel network using printed tiles on a 5×5 board.

The AI identifies the tile positions and orientations, reconstructs the network as a graph and verifies whether water can reach three agricultural destinations.

AI capabilities
- Visual board recognition
- Tile-position detection

- Graph construction
- Route validation
- Obstacle detection
- Automated scoring
- Animated, evidence-backed water-flow visualization

Rules
- Connected destination: +3
- Invalid outlet/leak: -1
- Time bonus: Up to +2
- Round duration: 90 seconds
- Tie-break: Shortest valid network, then time

Wow factor: Split-screen physical board and live digital reconstruction, with verified water flowing across the graph.

GAME 3: SOUQ SIGNAL

Mode: Individual competition.

Create an AI-observed shopping challenge inspired by UAE marketplaces.

Players receive a shopping mission and select physical product cards under a limited budget.

The cards represent products such as dates, coffee, spices, pottery and textiles.

Each card has:
- Unique visual marker
- Fixed price
- Product name
- Arabic/English labels

The player displays selected cards to the camera and verbally announces the products.

AI capabilities
- Visual item recognition
- Speech transcription
- Audio/video agreement checking
- Budget calculations
- Duplicate detection
- Rule validation
- Automated scoring

Rules
- Correct requested item: +2
- Valid matching spoken selection: +1
- Exceeded budget: -2
- Duplicate item: No additional item points
- Tie-break: Lower valid expenditure, then time

Wow factor: Live visual recognition, synchronized speech transcript, verified basket, remaining budget and explainable AI referee.

---

4. PREMIUM DESIGN SYSTEM

The application must look like a commercially polished, award-worthy product.

Visual identity

Colors:
- Midnight navy: #12243A
- Ivory: #F7F3EB
- Heritage teal: #167E80
- Emirates red: #B92D39
- Muted gold: #C6A36B
- Charcoal: #26313C

Use:
- Modern editorial typography
- Elegant spacing
- Strong typographic hierarchy
- Cinematic hero sections
- Interactive game cards
- Refined motion and microinteractions
- Premium scoreboards
- Responsive layouts
- Authentic UAE cultural references

Avoid generic templates, neon aesthetics, excessive glassmorphism, unreadable effects and unnecessary decorative complexity.

REQUIRED DESIGN REFERENCES

Research the following references before implementing the interface.

SPLINE — Interactive 3D

Spline for Web:
https://spline.design/web

Use this reference to understand how interactive 3D scenes integrate into React.

Reactive Orb — Spline Community:
https://app.spline.design/community/file/306ca2a5-d1ab-46ac-a27c-198575c82db0

Adapt its reactive-motion concept for WASL Arena's AI Referee visualization.

Create an original visual that reacts to analysis states:
- Idle
- Watching
- Processing
- Evidence verified
- Review required

Reeded Glass / Prism Hero Concept:
https://app.spline.design/community/file/a74713b7-6154-4b89-a5d9-47cfca3f413f

Study subtle depth, lighting and premium composition. Do not replicate its entire visual style.

Spline Community:
https://community.spline.design/

Explore suitable interactive elements for game selection and hero animations.

RECENT.DESIGN — Contemporary Interaction Design

Website Inspiration:
https://recent.design/websites

Study premium product layouts, transitions, navigation, composition and typography.

Krea Image Generation Loading State:
https://recent.design/i/f3cwpo5-ai-image-generation-loading-state

Reference the progressive reveal of AI-processing stages.

Apply the underlying interaction principle to WASL Arena's judging sequence:
Uploading → Extracting Evidence → Detecting Events → Applying Rules → Verified Result.

Krea Agent Onboarding:
https://recent.design/i/t865vsj-krea-agent-onboarding

Reference its structured onboarding and transitions for family registration, team setup and game configuration.

Linearity AI Launch Video:
https://recent.design/i/3qpnwhw-linearity-ai-launch

Study concise product storytelling, interface-focused motion and clean transitions for the competition video.

Design integration requirements

Do not copy complete interfaces, branding or protected design assets.

Use the references to inform original layouts, interactions and 3D elements.

Only embed a Spline scene if it is permitted for reuse and technically reliable.

If a specific scene is inaccessible or unsuitable, recreate the design principle using original React, CSS, SVG or Three.js implementations.

Optimize performance and provide a lightweight fallback if interactive 3D cannot load.

Do not let visual enhancements interfere with gameplay or judging reliability.

---

5. APPLICATION SCREENS

Develop eight complete screens:

1. **Launch:** Cinematic branding, interactive hero, championship introduction.
2. **Family Setup:** Players, teams, language, match configuration.
3. **Game Library:** Exactly three visually distinctive game cards.
4. **Game Preparation:** Instructions, equipment, rules, calibration.
5. **Live Arena:** Camera, recording, timer, rounds, scores.
6. **AI Referee Replay:** Evidence, detections, rules and verification.

7. **Family Cup:** Results, winners, MVP and standings.
8. **Evaluation Lab:** Actual test cases, outputs and limitations.

Every button, link and navigation action must work.

Provide downloadable printable game boards and cards for Falaj Flow and Souq Signal.

---

6. TECHNICAL IMPLEMENTATION

Frontend
- React
- TypeScript
- Vite
- Tailwind CSS
- Responsive routing
- MediaRecorder API
- Canvas/SVG annotations
- Spline integration where useful

Backend
- Python
- FastAPI
- OpenCV
- FFmpeg
- Appropriate pretrained vision models
- Speech transcription
- Pydantic schemas
- Game-specific deterministic rule engines

Gemini may support structured interpretation, transcription or explanations, but it must not invent events or serve as an unchecked scoring authority.

Processing pipeline

Gameplay → Video/Audio → Media Extraction → Visual/Audio Detection → Structured Events → Rule Validation → Evidence Report → Host Review → Verified Score.

Implement real processing for all three games.

Every major event must reference its underlying evidence.

Create:
- Source-video hashes
- Rule-version identifiers
- Event timestamps
- Structured judging reports
- Validation results
- Review states
- Session audit history

Uncertain detections must be labeled `NEEDS_REVIEW`.

Never hardcode fake positive results into the live judging workflow.

Testing

Include executable tests for:

Football
- Valid goal
- No-goal
- Uncertain detection
- Score update
- Tie-break

Falaj Flow
- Complete valid route
- Partial network
- Blocked route
- Invalid connection
- Correct scoring

Souq Signal
- Correct selection
- Incorrect selection
- Audio/video mismatch
- Budget violation
- Duplicate selection

Report only tests that actually ran.

Use authentic recorded gameplay for at least one end-to-end demonstration per game.

---

7. GITHUB AND VERCEL

Create a standalone repository named:

`wasl-arena`

Do not modify DOMINO.

Organize frontend, backend, tests, documentation, presentation assets, video assets and AI-usage records.

Prepare:

- README.md
- package.json
- requirements.txt
- .env.example
- .gitignore
- vercel.json where needed
- Deployment configuration
- Backend setup guide
- Test instructions

The React frontend must deploy through GitHub → Vercel.

Use Render or another appropriate backend host for video processing.

Configure the backend URL through environment variables.

Keep all credentials server-side.

Ensure:
- Successful production build
- Functional routes
- Responsive design
- Working API connection
- No missing assets
- No exposed secrets

Do not claim that publication is complete until the live application is verified.

---

8. SIX-PAGE TECHNICAL REPORT

Generate the complete six-page technical report.

Output:
- Editable Word document
- Final PDF

Use a professional academic design, with consistent navy/teal accents and real application screenshots.

Page 1: Executive Summary, Problem and Objectives

Page 2: System Concept, Three Games and User Journey

Page 3: System Architecture, Technology and AI Pipeline

Page 4: Football, Falaj and Souq AI Methodologies

Page 5: Testing, Results, Validation and Limitations

Page 6: Fairness, Privacy, Deployment, Impact and Conclusions

Include diagrams, tables, references and only genuine technical results.

Exactly six pages.

---

9. THIRTY-SLIDE PRESENTATION

Generate an editable PowerPoint with exactly 30 slides.

Use the WASL Arena design language and genuine screenshots.

Slide sequence:

1. Title
2. Vision
3. Problem
4. Why Family Gaming Matters
5. Official Challenge
6. Our Solution
7. Value Proposition
8. Signature Innovation
9. User Journey
10. Three-Game Collection
11. Football Overview
12. Football AI Tracking
13. Football Referee Replay
14. Falaj Flow Overview
15. Falaj Graph Intelligence
16. Falaj Scoring
17. Souq Signal Overview
18. Multimodal AI
19. Souq Transaction Verification
20. Family Host Console
21. AI Referee Replay
22. Technical Architecture
23. AI Models and Algorithms
24. Evidence Integrity
25. Fairness and Privacy
26. Evaluation Methodology
27. Actual Test Results
28. GitHub and Vercel Deployment
29. Limitations and Future Development
30. Conclusion and Final Impact

Make every slide visually cohesive, concise and technically meaningful.

Include presenter notes.

Do not fabricate evaluation data.

---

10. TWO-MINUTE VIDEO — 24 CLIPS

Develop a complete cinematic video production package.

Maximum length: 120 seconds.

Clip length: Exactly five seconds per clip.

Total: 24 clips × 5 seconds.

Format: 16:9, 1080p MP4.

Transitions must be included within the clip durations. Do not add extra runtime.

Required storyboard

Clip	Content
01	Cinematic family-gaming opening
02	The problem of fair game judging
03	WASL Arena brand reveal
04	Three games, one AI referee
05	Actual app landing page
06	Family registration
07	Three-game library
08	Host console
09	Real football gameplay
10	Football video analysis
11	Player and ball tracking
12	Goal verification and score
13	Falaj physical gameplay
14	Visual tile recognition
15	Digital irrigation graph
16	Verified Falaj scoring
17	Souq physical gameplay
18	Visual and speech recognition
19	Budget and item verification
20	Souq results
21	AI Referee Replay
22	Testing and technical evidence
23	Family Cup leaderboard
24	Cinematic closing statement

For each clip, produce:

1. Exact five-second scene description.
2. Detailed AI-video-generation prompt.
3. Camera framing and movement.
4. Source asset or real app screenshot.
5. Exact narration.
6. On-screen text.
7. Sound effects/music guidance.
8. Transition to next clip.

Mandatory video rules

- Use actual screenshots captured from the completed application.
- Use genuine gameplay footage for scoring demonstrations.
- Do not fabricate functioning UI or AI evidence.
- Keep narration short enough for five seconds.
- Use consistent colors, typography and atmosphere.
- Make the video emotionally engaging and technically credible.
- Generate a full narration transcript.
- Generate a shot-by-shot editing timeline.
- Provide an asset checklist.
- Provide final export settings.

Use the Recent.design launch-video reference to guide the visual pacing, but create an original film.

If direct MP4 production is unavailable, generate a complete production-ready storyboard, prompts, narration, timing sheet and editing instructions.

---

11. AI CHAT USAGE REPORT

Prepare a single PDF documenting authentic AI assistance throughout development.

Include:
- AI tools and models
- Development phases
- Architecture prompts
- Code generation
- Debugging
- Testing
- Design iterations
- Documentation creation
- Presentation development
- Video scripting
- Human verification and adjustments

Import original conversation exports where provided.

Do not invent chat transcripts.

If exports are unavailable, identify them as pending and provide a clear compilation process.

Redact API credentials and sensitive information.

The report must accurately represent the team's actual AI usage.

---

12. EXECUTION PRIORITIES

Follow this strict order.

Priority 1 — Prototype
Build all three games, real judging adapters, host controls and evidence replay.

Priority 2 — Technical verification
Test the application and collect genuine screenshots, recordings and results.

Priority 3 — GitHub/Vercel
Prepare and verify the production deployment.

Priority 4 — Technical report
Generate the six-page document from actual implementation evidence.

Priority 5 — Presentation
Generate thirty editable slides using genuine prototype results.

Priority 6 — Video
Produce the twenty-four five-second clip prompts, narration, screenshots and editing package.

Priority 7 — AI chat report
Compile authentic AI conversations into one PDF.

Priority 8 — Final submission audit
Verify every deliverable against the official brief.

Work autonomously and minimize interruptions.

If external authorization or credentials are required, identify the blocker and continue everything that can be completed safely.

---

13. NON-NEGOTIABLE FINAL QUALITY CHECKS

Before declaring completion, verify:

- Exactly three games.
- Football included and functional.
- Two original games implemented.
- Both team and individual play.
- Real video/audio-based AI judging.
- Evidence-backed scoring.
- Functional host controls.
- Complete navigation.
- Actual test results.
- No fabricated demonstrations.
- No exposed credentials.
- Working production build.
- GitHub/Vercel deployment prepared and verified where access permits.
- Six-page technical report.
- Thirty-slide presentation.
- Authentic AI chat PDF.
- Twenty-four five-second video clips or complete production package.
- Total video runtime no more than 120 seconds.

Begin implementation immediately. Build the prototype first, then generate every submission artifact from the verified application.

FINAL OBJECTIVE: Deliver a distinctive, polished, fully functional and technically defensible Cycle 3 competition entry, demonstrating that AI can make family gaming more interactive, inclusive, culturally meaningful and transparently fair.

Sources: attached + the specific link for the theme
https://docs.google.com/document/u/0/d/1C87Qkqq5I2An8QM95JX0wB5OQ_ipm1DFmOR45PQE-xQ/mobilebasic#

Spline / Recent resources

Spline — Interactive 3D Design

1. [Spline for Web — Interactive 3D Integration](https://spline.design/web?utm_source=chatgpt.com)
2. [Reactive 3D Orb — AI Referee Animation](https://app.spline.design/community/file/306ca2a5-d1ab-46ac-a27c-198575c82db0?utm_source=chatgpt.com)
3. [Reeded Glass / Prism Hero Concept](https://app.spline.design/community/file/a74713b7-6154-4b89-a5d9-47cfca3f413f?utm_source=chatgpt.com)

4. [Spline Community — 3D Templates and Interactions](https://community.spline.design/?utm_source=chatgpt.com)

Recent.design — UI/UX and Motion Inspiration

1. [Recent.design — Website Inspiration](https://recent.design/websites?utm_source=chatgpt.com)
2. [Krea AI — Animated Loading States](https://recent.design/i/f3cwpo5-ai-image-generation-loading-state?utm_source=chatgpt.com)
3. [Krea Agent — Interactive Onboarding](https://recent.design/i/t865vsj-krea-agent-onboarding?utm_source=chatgpt.com)
4. [Linearity AI — Cinematic Launch Video](https://recent.design/i/3qpnwhw-linearity-ai-launch?utm_source=chatgpt.com)

My top three for WASL Arena:

* Spline Reactive Orb: For the AI Referee's interactive visualization.
* Krea AI Loading States: For video analysis, event detection and score verification.
* Recent.design Website Gallery: For the landing page, game-selection cards and overall interface design.

Also this is the google studies API if needed

[REDACTED-API-KEY]
WASL Arena Key
projects/270696470798
270696470798

USER · 2026-10-29 11:20:56 UTC

[Image: source: C:\Users\aaa\AppData\Local\Temp\claude\C--Users-aaa-AppData-Local-Claude-Data-claude-pzs8sxrjxfjjc-scratch-workspaces-e54784a8-37a4-4cc2-9d86-5fb6e5584d2c-35552a5f-98f9-4e0e-b06f-d1a96e9ca10a-scratch-2026-10-29-38eede\b39dc4d9-905e-4bd9-9e8e-12e589894be4\images\1.jpg]

CLAUDE · 2026-10-29 11:21:00 UTC

I'll start by reading the official brief, then build the prototype before any deliverables.

CLAUDE · 2026-10-29 11:21:17 UTC

No Python, no ffmpeg, and the Read tool can't render PDFs here. I'll try the PDF tools and check what else is installable.

CLAUDE · 2026-10-29 11:21:31 UTC

The brief PDF has no text layer (it's image-based). I'll check the other one and try the PDF viewer.

CLAUDE · 2026-10-29 11:21:48 UTC

The brief PDF is just an invitation email (logos, no problem text). The real problem statement is the Google Doc you linked, so I'll fetch that.

CLAUDE · 2026-10-29 11:21:58 UTC

The document body is three embedded images. I'll try to pull them directly.

CLAUDE · 2026-10-29 11:22:07 UTC

Downloading the three pages at higher resolution, then reading them.

CLAUDE · 2026-10-29 11:23:45 UTC

Toolchain is workable. I'm scaffolding the repo, creating a Python venv and installing the backend dependencies in the background while I scaffold the frontend.

CLAUDE · 2026-10-29 11:24:29 UTC

The share page is JS-rendered, so I'll read it in the built-in browser while I scaffold the frontend.

CLAUDE · 2026-10-29 11:25:57 UTC

The conversation rendered; extracting all turns from the DOM.

CLAUDE · 2026-10-29 11:26:05 UTC

Saving the full ChatGPT transcript verbatim into the repo via a local one-shot receiver (avoids routing 47KB through my context).

CLAUDE · 2026-10-29 11:26:18 UTC

The browser blocks posts to localhost. Trying the clipboard route instead.

CLAUDE · 2026-10-29 11:29:21 UTC

The Gemini key works (`gemini-2.5-flash` is available) and stays in a gitignored `.env`. Writing the core modules and rule engines now.

CLAUDE · 2026-10-29 11:31:07 UTC

Now the vision layer: video I/O + ffmpeg, ArUco marker utilities, the football tracker/goal/possession detector, and the speech adapter.

CLAUDE · 2026-10-29 11:31:39 UTC

Now the football judge (colour-segmentation tracking, goal-zone detection, conservative possession).

CLAUDE · 2026-10-29 11:32:34 UTC

Now the Falaj Flow reader (ArUco board homography → tile type/rotation → graph) and the Souq Signal judge.

CLAUDE · 2026-10-29 11:34:22 UTC

Now synthetic test-fixture generators (clearly labelled as synthetic, not real gameplay) so every pipeline can be tested with known ground truth.

CLAUDE · 2026-10-29 11:35:13 UTC

Generating the fixtures (includes synthetic TTS audio) while I write the API layer.

CLAUDE · 2026-10-29 11:36:11 UTC

Now the executable tests (rule engines, pipelines on fixtures, integrity/API).

CLAUDE · 2026-10-29 11:37:36 UTC

Two genuine bugs caught by the tests: the well's own inlet was counted as a leak, and Arabic words with the "ال" prefix (التمر) weren't matched. Fixing both.

CLAUDE · 2026-10-29 11:38:03 UTC

Pipeline tests on all three games pass; the one failure is real: a host-accepted "possible goal" wasn't being counted. Fixing the rule.

CLAUDE · 2026-10-29 11:40:47 UTC

Backend: **46 tests passing, 0 failing**. Now the frontend: infra, state, API client and the AI-Referee orb first.

CLAUDE · 2026-10-29 11:41:34 UTC

Now the Orb (original Three.js reactive orb with CSS fallback), layout/nav, judging sequence and Falaj digital twin.

CLAUDE · 2026-10-29 11:42:43 UTC

Writing the sample/evaluation exporter (runs the real pipelines on the fixtures and records actual vs expected), then launching it in the background.

CLAUDE · 2026-10-29 11:43:11 UTC

Now App routing plus the first four screens (Launch, Family Setup, Game Library, Game Preparation).

CLAUDE · 2026-10-29 11:44:23 UTC

All 9 fixtures match ground truth and pytest is 46/46. The Gemini transcription step came back UNAVAILABLE, so I'm checking why.

CLAUDE · 2026-10-29 11:48:14 UTC

While that model check runs in the background, I'm writing the Preparation and Live Arena screens.

CLAUDE · 2026-10-29 11:49:02 UTC

Transcription works with newer models: `gemini-3.1-flash-lite` returns correct text with timestamps within ~0.3 s. Switching the default and re-running the exporter in the background.

CLAUDE · 2026-10-29 11:50:03 UTC

Now the key screen: the AI Referee Replay (video + synced overlays, evidence panel, host review, integrity).

CLAUDE · 2026-10-29 11:51:28 UTC

Now Family Cup and Evaluation Lab.

CLAUDE · 2026-10-29 11:52:07 UTC

Now the printables generator (board, tiles, cards with real ArUco markers), then build the frontend.

CLAUDE · 2026-10-29 11:52:39 UTC

Type-check is clean. Building, and starting the backend so I can check the real UI.

CLAUDE · 2026-10-29 11:53:06 UTC

Build passes and the real speech check now works: Gemini transcribed the synthetic audio and the deterministic engine scored it 9/9 as expected. Now visually checking the screens.

CLAUDE · 2026-10-29 11:53:28 UTC

Orb renders. One bug: stat numbers are invisible on the dark hero because `.display` forces navy text. Fixing, then checking the Replay screen at desktop size.

CLAUDE · 2026-10-29 11:53:47 UTC

Replay works end-to-end: synced overlay, goal zones, evidence frames, rule text, host actions. I need real screenshot files for the report/deck, so I'll capture them with headless Edge.

CLAUDE · 2026-10-29 12:00:11 UTC

Word is installed, so I can convert the report to PDF and verify the exact page count. Now grabbing the ChatGPT transcript via headless Chrome (no token cost) and committing the repo locally.

CLAUDE · 2026-10-29 12:00:43 UTC

Got all 6 ChatGPT turns. Parsing them into the repo as the authentic export.

CLAUDE · 2026-10-29 12:01:11 UTC

All 6 ChatGPT turns saved verbatim. Now the supporting repo files, then the six-page technical report generator (docx → PDF via Word with page-count check).

CLAUDE · 2026-10-29 12:03:06 UTC

Exactly 6 pages on the first build. Let me visually check the pages.

CLAUDE · 2026-10-29 12:03:46 UTC

Pages look right but there's spare room, so I'll enlarge the body text and re-check the page count.

CLAUDE · 2026-10-29 12:04:29 UTC

Report is solid (6 pages). I'll fix the clipped pipeline diagram, then build the 30-slide deck.

CLAUDE · 2026-10-29 12:06:10 UTC

Noted: I'll add the team details (members, team 204, code AYUCK, Cycle 3 track) to the prototype and every document. Starting with the app and the deck/report builders.

CLAUDE · 2026-10-29 12:07:57 UTC

Deck built (30 slides). Rendering it through PowerPoint to check the layout visually.

CLAUDE · 2026-10-29 12:08:37 UTC

The deck renders well (one title-slide text overlap to fix). Now the 24-clip video package, plus a real draft MP4 animatic assembled from actual screenshots and scratch narration.

CLAUDE · 2026-10-29 12:09:51 UTC

While the video renders, I'm exporting this Claude session's transcript for the AI-usage report (it will be key-redacted).

CLAUDE · 2026-10-29 12:10:03 UTC

That guard blocked my `rm -rf *`, which was unnecessary. I'm not retrying it. Extracting into a fresh directory instead.

CLAUDE · 2026-10-29 12:10:19 UTC

Transcript captured. Building the AI-usage report (team details, tools/models, phases, verification, then the verbatim ChatGPT and Claude Code conversations with the API key redacted).

CLAUDE · 2026-10-29 12:11:54 UTC

AI-usage PDF built (80 pages, key redacted). The video came out 120.05 s, so I'm trimming it to exactly 120.0 s and scanning the repo for secrets.

CLAUDE · 2026-10-29 12:12:20 UTC

Video is 24 clips of 5 s each; I'll re-encode the tail to land at ≤120.0 s. Then README, final test run, build, commit, and GitHub.

CLAUDE · 2026-10-29 12:13:34 UTC

46/46 tests pass and the production build is clean. Committing and creating the GitHub repository (private; you can make it public later).

CLAUDE · 2026-10-29 12:14:22 UTC

Pushed. Cleaning up the dev servers and browser viewport I started.

CLAUDE · 2026-10-29 12:14:42 UTC

WASL Arena is built and pushed to a private GitHub repo, but it is not deployed, and no real gameplay footage has been judged yet. All deliverables are in the repo.

Repo: https://github.com/MayMourad/wasl-arena (private, under the logged-in `gh` account `MayMourad`; make it public if the submission needs that). Local copy is in `Desktop\6th Annual ADU CoE STEM Hackathon\wasl-arena`.

What's done

- **Prototype:** all 8 screens work.
  - Three games: Football AI, Falaj Flow (team) and Souq Signal (individual).
  - The Live Arena records from camera and microphone or takes an upload.
  - Replay shows the video with synced overlays, the evidence frames and the rule applied. The host can accept or reject any call, and the score is recomputed.
  - Printable boards, tiles and product cards are included.
- **Tests:** 46/46 pass in the final run, and 9/9 synthetic fixtures match their ground truth.
- **Speech:** a live Gemini transcription call returned the correct words for the Souq test audio and scored 9/9. Gemini only transcribes, and the rules decide the score.
- **Bugs the tests caught:**
  - the well's own inlet was counted as a leak;
  - Arabic words with the "ال" prefix weren't matched;
  - an accepted "possible goal" wasn't scored.

- The originally planned Gemini model is retired for new users, so I switched to `gemini-3.1-flash-lite`.
- **Documents:** the technical report is exactly 6 pages (Word and PDF, page count checked in Word). The deck is 30 editable slides with presenter notes. The AI usage report is a PDF with your ChatGPT chat and this Claude session included verbatim, with the API key redacted.
- **Video:** the full production package covers all 24 clips of 5 s, and the draft animatic is 120.0 s. It uses real app screenshots and scratch text-to-speech narration.
- **Team details:** team 204, code AYUCK, Cycle 3 and all three members appear in the app footer, the launch page, printables, report, deck, video credits, AI report and README.

What you still need to do

1. **Record real gameplay.** Every test clip is a synthetic render, and the report, deck and Evaluation Lab say so. Clips 09, 13 and 17 in the video are marked with a red banner and need real footage. I could not film anything, so no accuracy figure is claimed.
2. **Deploy.** The frontend goes to Vercel with Root Directory `frontend` and `VITE_API_URL` set. The backend goes to Render via `render.yaml`. I have no login for either, so no live URL exists, and live judging needs the backend deployed. The pre-computed sample replays work without it.
3. **Rotate the Gemini key.** It was pasted in chat, so it is also in the session export zip I saved to your Downloads folder (`session-export-1793275792519.zip`). It is not in the repo.
4. **Check the files before submitting.** The first pages of the report, the title and closing slides, and the video's red-banner clips are worth a look.

USER · 2026-10-29 12:14:48 UTC

Walk me through

CLAUDE · 2026-10-29 12:14:59 UTC

Deploy the backend first, because Vercel needs its URL. Both services need GitHub access to the private `MayMourad/wasl-arena` repo, and you need to be logged in to both.

1. Backend on Render (about 5 min)

1. Go to render.com, choose **New → Blueprint**, connect GitHub and pick `wasl-arena`. It reads `render.yaml` and creates the `wasl-arena-judge` service. If you pick "New → Web Service" instead, use **Root Directory** `backend`, **Build** `pip install -r requirements.txt`, **Start** `uvicorn app.main:app --host 0.0.0.0 --port $PORT`.
2. When asked for environment values:
   - `GEMINI_API_KEY`: use a **new** key from Google AI Studio, not the one pasted in chat.
   - `JUDGE_SIGNING_KEY`: Render generates this for you.
   - `CORS_ORIGINS`: put `*` for now and tighten it in step 4.
3. After it deploys, open `https://<your-service>.onrender.com/api/health`. It should return `"status":"ok"` and `"transcription":"configured"`.

Free-tier limits:
- The service sleeps after idle, so the first request is slow. Open `/api/health` a few minutes before you demo.
- The disk is wiped on restart, so reports and uploads don't persist.
- 512 MB of RAM may be tight for long videos. Use the Starter plan if a judging job fails, and keep clips short (under 60 s).

2. Frontend on Vercel (about 3 min)

1. At vercel.com, choose **Add New → Project** and import `wasl-arena`.
2. Set **Root Directory** to `frontend`. The framework is detected as Vite, and `vercel.json` handles the rest.
3. Under **Environment Variables**, add `VITE_API_URL` = `https://<your-service>.onrender.com` with no trailing slash.
4. Click **Deploy**.

3. Verify (this is what makes it "published")

On the Vercel URL, check these:
- The header badge says **"AI judge online"**.
- `/replay?report=sample-football_goal` plays the video with overlays.

- Live Arena → upload one of the clips from `fixtures/` → the stages run → Replay opens. Regenerate them with `python tools/make_fixtures.py`, since the fixture mp4s aren't in git.
- Open the `/printables/falaj-board.html` page and refresh `/cup` and `/lab` to confirm routing works.

4. Lock it down

In Render, set `CORS_ORIGINS=https://<your-app>.vercel.app` and redeploy.

Then send me both URLs and I'll add them to the README and check them. If you're pressed for time, deploying the frontend alone still gives a working site with all the sample replays. Only live judging needs the backend.

USER · 2026-10-29 12:16:32 UTC

Specifically what should i name the blueprint, and which new key are you talking about? DO i need to create a new wasl project? Afterwards I will publish on vercel.

CLAUDE · 2026-10-29 12:16:41 UTC

Blueprint name: it can be anything, for example `wasl-arena`. It's only a label in Render's dashboard. The actual service name, `wasl-arena-judge`, comes from `render.yaml`, and that name sets your URL (`https://wasl-arena-judge.onrender.com`, or with a suffix if the name is taken).

New Render project: you don't need one. If Render shows a "Project" dropdown, you can leave the default or create one called "WASL Arena". It makes no difference to the deployment.

The new key: this is a new Gemini API key, used only for speech transcription in Souq Signal. Your current key was pasted into our chat, so it now sits in the session export in your Downloads folder.
1. Go to https://aistudio.google.com/apikey and click **Create API key**. Using your existing WASL Arena project is fine.
2. Paste the new key directly into Render's `GEMINI_API_KEY` field. Don't send it to me.
3. After the competition, delete the old key on that same page.

If you're short on time, you can paste the old key into Render now and rotate it after you submit. The risk is only that it stays exposed until you rotate it.

Render will ask you for two values during the Blueprint step, and you can leave the third alone:
- `GEMINI_API_KEY`: the new key.
- `CORS_ORIGINS`: type `*` for now. Replace it with your Vercel URL once you have one.
- `JUDGE_SIGNING_KEY`: Render generates this itself.

Once the Render service shows "Live", open `/api/health` on it. After that, go ahead with the Vercel steps from before and send me both URLs.

USER · 2026-10-29 12:21:57 UTC

deploy failed, find out why. https://dashboard.render.com/web/srv-dbhjkf7b9u1s73f1cscg/deploys/dep-dbhjkffb9u1s73f1cth0

CLAUDE · 2026-10-29 12:22:01 UTC

I'll open the deploy page and read the log.

CLAUDE · 2026-10-29 12:22:26 UTC

I can't see why it failed, because the Render dashboard sits behind a login and my built-in browser isn't signed in. I won't sign in to Render for you.

You can give me the error in either of two ways:

1. **Paste the log.** On that deploy page, copy the last 30–40 lines of the log (the part with the red error) and paste them here. This is fastest.
2. **Let me read it in your Chrome.** Your own Chrome is already logged in to Render. If you say "use my Chrome", I'll read the log there.

I'd only read the log, not change any settings.

I can't tell the cause from here. If you want to try something first, the usual cause of a Python build failing on Render is the Python version. You can set an environment variable `PYTHON_VERSION=3.12.8` on the service and redeploy. I can also add that to `render.yaml` and push it, but I'd rather confirm from the log than guess.

USER · 2026-10-29 12:22:51 UTC

use my chrome, you have full access to do as you wish and want.

CLAUDE · 2026-10-29 12:23:10 UTC

Still trying to open the Render deploy log in your Chrome; the first attempt got bounced by a domain-change check, so I'm retrying with the page load and read in one step.

CLAUDE · 2026-10-29 12:23:25 UTC

Your Chrome is also showing Render's login page, so it isn't signed in either. I won't sign in myself, and I haven't entered any credentials.

Choose one:

1. **Sign in yourself.** I left a Render tab open in the Chrome window I control. Sign in there (GitHub is easiest), then say "signed in" and I'll read the log right away.
2. **Paste the log.** On the deploy page, copy the last 30–40 lines (the part with the error) and paste them here.

USER · 2026-10-29 12:24:22 UTC

==> Cloning from https://github.com/MayMourad/wasl-arena
==> Checking out commit c0060d42a7a3ab555e3dd2f54cadcba053c16a2a in branch main
==> Using Python version 3.14.8 (default)
==> Docs on specifying a Python version: https://render.com/docs/python-version
==> Installing Python version 3.14.8...
==> Using Poetry version 2.1.3 (default)
==> Docs on specifying a Poetry version: https://render.com/docs/poetry-version
==> Running build command 'pip install -r requirements.txt'...

[notice] A new release of pip is available: 26.2.1 -> 26.2.2
[notice] To update, run: pip install --upgrade pip
ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'
==> Build failed 😞
==> Common ways to troubleshoot your deploy: https://render.com/docs/troubleshooting-deploys

CLAUDE · 2026-10-29 12:24:29 UTC

Cause: the service is building from the repo root, where there is no `requirements.txt`. That file is in `backend/`. This means the **Root Directory** setting is empty. It was probably created as a plain Web Service, not from the blueprint, so `rootDir: backend` in `render.yaml` was never applied. Render's Python 3.14.8 is not the problem: the same version installed these dependencies fine for me locally.

Fix, in Render (about 1 minute):
1. Open your service → **Settings**.
2. Set **Root Directory** to `backend`.
3. Check these two fields on the same page:
   - **Build Command:** `pip install -r requirements.txt`
   - **Start Command:** `uvicorn app.main:app --host 0.0.0.0 --port $PORT`
4. Click **Save Changes**, then **Manual Deploy → Deploy latest commit**.

Under **Environment**, `GEMINI_API_KEY` (your new key), `JUDGE_SIGNING_KEY` (any long random string) and `CORS_ORIGINS` (`*` for now) must also be set. If the service wasn't made from the blueprint, they may be missing.

When it shows "Live", open `https://<your-service>.onrender.com/api/health`. If the next deploy fails, paste the new log and I'll look at it.

USER · 2026-10-29 12:27:31 UTC

Here is the service id
srv-dbhjkf7b9u1s73f1cscg

LIVE.

CLAUDE · 2026-10-29 12:27:32 UTC

Checking the health endpoint.

CLAUDE · 2026-10-29 12:27:42 UTC

The backend is live at **https://wasl-arena.onrender.com**. Before you move on, I'll run a real judging job against it to check it can process video.

CLAUDE · 2026-10-29 12:28:42 UTC

The backend is live at **https://wasl-arena.onrender.com**. I ran a real Falaj judging job on it end to end and it scored 11, matching the local result. The report's integrity check passed. The job took about 40 s on the free tier.

One problem remains: the health check says `"transcription":"not configured"`. `GEMINI_API_KEY` isn't set on the Render service, so Souq Signal's spoken-match points would stay NEEDS_REVIEW. In Render, go to **Environment**, add `GEMINI_API_KEY` with your new key and `GEMINI_MODEL=gemini-3.1-flash-lite`, then save (it redeploys). Check `/api/health` shows `"configured"` afterwards.

Next, on Vercel:
1. Import `MayMourad/wasl-arena`.
2. Set **Root Directory** to `frontend`.
3. Add the environment variable `VITE_API_URL` = `https://wasl-arena.onrender.com`, with no trailing slash.
4. Deploy.

Then send me the Vercel URL. I'll check the site loads and the "AI judge online" badge shows, and tell you what to put in Render's `CORS_ORIGINS` (replacing `*`).

Fix the Gemini key first if you can, because a Souq test on the live site depends on it.

USER · 2026-10-29 12:35:37 UTC

Done, on vercel I launched the front end and backend, check my vercel.

Front end: https://frontend-three-fawn-47.vercel.app/
back end:
https://vercel.com/mam-e31b/backend/5c5Ju6kN2EAZrfKoQgv9ePAxNZnq

What's next?

CLAUDE · 2026-10-29 12:35:43 UTC

Checking the live frontend now.

CLAUDE · 2026-10-29 12:36:06 UTC

The frontend is up and its routes, sample replays, printables and evaluation page all load. Transcription is now "configured" on Render too. The remaining problem is that the frontend isn't talking to the Render backend.

What I found: the deployed JavaScript doesn't contain `https://wasl-arena.onrender.com`. The live app is calling `/api/...` on its own Vercel domain, which returns 404, so the header badge will show **"AI judge offline"** and uploads will fail. Vite writes `VITE_API_URL` into the build, so it must be set **before** the build runs.

What to do next:

1. In Vercel, open the **frontend** project (`frontend-three-fawn-47`) → **Settings → Environment Variables**.
2. Add `VITE_API_URL` = `https://wasl-arena.onrender.com`, with no trailing slash, for Production, Preview and Development.
3. Go to **Deployments**, open the latest one, and choose **Redeploy** (untick "use existing build cache").
4. Tell me when it's done and I'll check the bundle and the live site.

Afterwards, in Render, change `CORS_ORIGINS` from `*` to `https://frontend-three-fawn-47.vercel.app`.

The Vercel "backend" project: you don't need it. The real backend is the one on Render, because Vercel's serverless functions can't run the video processing (FFmpeg and OpenCV jobs). I can't open that link without a login. I'd leave the project in place but unused, or delete it so nobody mistakes it for the judge. For the submission, give the frontend URL and mention the Render API.

USER · 2026-10-29 12:38:40 UTC

Where can I find the cors origins in render? Also, I re deployed the front end with the upated environemntal variable

CLAUDE · 2026-10-29 12:38:49 UTC

The new build now points at the Render backend. Checking the live site in a browser.

CLAUDE · 2026-10-29 12:39:02 UTC

The live site now shows **"AI judge online"**, so Vercel is talking to your Render backend.

Where to find CORS in Render: it's an environment variable.

1. Open your Render service (`wasl-arena`).
2. In the left sidebar, click **Environment**.
3. Find `CORS_ORIGINS`. If it isn't there, click **Add Environment Variable** and name it exactly that.
4. Set the value to `https://frontend-three-fawn-47.vercel.app` (no trailing slash, no spaces).
5. Click **Save, rebuild and deploy**. It takes a couple of minutes, then the service is live again.

Until you do this, `CORS_ORIGINS=*` lets any website call your judge. Nothing breaks if you leave it for now, but it's worth tightening before you submit.

After the redeploy, do a full live test: Live Arena → upload a clip → check the stages run and Replay opens. The Render free tier sleeps when idle, so the first request after a break can take about 30 s.

Two things I haven't done:

1. **README.** I'll add the live URLs once you confirm the CORS change, so I can verify the site afterwards.
2. **Real footage.** That is still the biggest gap. Record short real clips (a football goal, a Falaj board, a Souq round) and upload them through the live site. Send me the results and I'll update the report, deck and video package.

USER · 2026-10-29 12:40:30 UTC

Both re deployments done, Now:
Let's work on finalizing the script, include image generation prompts, animation prompts, and voiceover. 2 minutes.
Afterwards, share the script as a downloadable pdf file, and then work on the technical pdf and then the 30-slide presentation, finally followed by the AI usage file as per the requirements.

CLAUDE · 2026-10-29 12:41:00 UTC

Starting with the finished script (image prompts, animation prompts, voiceover) as a PDF, then I'll refresh the report, deck and AI-usage file with the live deployment details. First a quick check that your redeployments took effect.

CLAUDE · 2026-10-29 12:41:10 UTC

Deployment is verified: CORS now allows only your Vercel site, the backend is healthy, and transcription is configured. Now the final script content.

CLAUDE · 2026-10-29 12:42:01 UTC

All 24 voiceover lines are 12 words or fewer, so they fit 5 seconds each. Now the script PDF builder, and I'm re-rendering the animatic with

the final voiceover in the background.

CLAUDE · 2026-10-29 12:43:15 UTC

Two small fixes: stray asterisks in a table cell, and one awkward phrase in the clip 02 prompt. I'll also stop clip cards splitting across pages.

CLAUDE · 2026-10-29 12:43:39 UTC

The script PDF is done (9 pages). Trimming the rebuilt animatic to ≤120 s, then sending you the script.

CLAUDE · 2026-10-29 12:44:20 UTC

Now the technical report: the deployment section gets the real URLs and what I actually verified.

CLAUDE · 2026-10-29 12:44:49 UTC

The report is clean apart from stray backtick characters, which I'm removing. Updating the deck's deployment slide at the same time.

CLAUDE · 2026-10-29 12:45:25 UTC

One overlap on slide 13 (captions under the evidence frames collide with the bullets). Fixing it, then refreshing the AI-usage export with everything since the last one.